

GDI BITNE FUNKCIJE

```
void CMy18724FilipLakicevicView::DegToRad(float deg) { return deg *  
3.14f / 180.0f; }
```

```
void CMy18724FilipLakicevicView::Translate(CDC* pDC, float dx, float  
dy, bool rightMultiply)  
{  
    XFORM xf { 1, 0, 0, 1, dx, dy };  
    pDC->ModifyWorldTransform(&xf, rightMultiply ?  
        MWT_RIGHTMULTIPLY : MWT_LEFTMULTIPLY);  
}
```

```
void CMy18724FilipLakicevicView::Rotate(CDC* pDC, float angleDeg,  
bool rightMultiply)  
{  
    angle = DegToRad(angle);  
  
    XFORM xform;  
    xform.eM11 = cos(angle);  
    xform.eM12 = sin(angle);  
    xform.eM21 = -sin(angle);  
    xform.eM22 = cos(angle);  
    xform.eDx = 0.0;  
    xform.eDy = 0.0;  
  
    pDC->ModifyWorldTransform(&xform, rightMultiply ?  
        MWT_RIGHTMULTIPLY : MWT_LEFTMULTIPLY);  
}
```

```
void CMy18724FilipLakicevicView::OnKeyDown(UINT nChar, UINT  
nRepCnt, UINT nFlags)  
{  
    Break or if's  
  
    Invalidate();  
}
```

```
void CMy18724FilipLakicevicView::Scale(CDC* pDC, float sX, float sY,  
bool rightMultiply)
```

```
{  
    XFORM xform{};  
    xform.eM11 = sX;  
    xform.eM12 = 0.0;  
    xform.eM21 = 0.0;  
    xform.eM22 = sY;  
    xform.eDx = 0.0;  
    xform.eDy = 0.0;  
  
    pDC->ModifyWorldTransform(&xform, rightMultiply ?  
MWT_RIGHTMULTIPLY : MWT_LEFTMULTIPLY);  
}
```

```
bool CMy18724FilipLakicevicView::OnEraseBkgnd(CDC* pDC)
```

```
{  
    return true;  
}
```

```
void CMy18724FilipLakicevicView::Mirror(CDC* pDC, bool mx, bool my,  
bool rightMultiply)
```

```
{  
    Scale(pDC, mx ? -1 : 1, my ? -1 : 1, rightMultiply);  
}
```

Inicijalizacija metafile

```
GetEnhMetaFile(CString("GDI2018Resursi\\viljuska.emf"));
```

Brisanje metafile

```
DeleteEnhMetaFile(img4);
```

```
// Crtanje pozadinske slike u prirodnoj velicini
void CMy18724FilipLakicevicView::DrawBackground(CDC* pDC)
{
    CRect client;
    GetClientRect(&client);
    pozadina->Draw(pDC, CRect(0, 0, pozadina->Width(),
    pozadina->Height()), client);
}
```

```
/* Crtanje pozadinske slike tako da
je uvek centrirana po X-osi, prilikom promene veličine prozora, a donja
ivica se poklapa sa
donjom ivicom klijentskog dela prozora */
void CMy18724FilipLakicevicView::DrawBackground(CDC* pDC) {
    CRect client;
    GetClientRect(&client);
    int x, y, w, h;
    x = (client.Width() - pozadina->Width()) / 2;
    w = x + pozadina->Width();
    y = client.Height() - pozadina->Height();
    h = y + pozadina->Height();
    pozadina->Draw(pDC, CRect(0, 0, pozadina->Width(),
    pozadina->Height()), CRect(x, y, w, h));
}
```

```

void CMy18724FilipLakicevicView::DrawImgTransparent(CDC* pDC,
DImage* plmage)
{
    int w = plmage->Width();
    int h = plmage->Height();

    CBitmap bmpImage, bmpMask;
    bmpImage.CreateCompatibleBitmap(pDC, w, h);
    bmpMask.CreateBitmap(w, h, 1, 1, NULL);

    CDC* pSrcDC = new CDC();
    pSrcDC->CreateCompatibleDC(pDC);
    pSrcDC->SelectObject(bmpImage);

    CDC* pMaskDC = new CDC();
    pMaskDC->CreateCompatibleDC(pDC);
    pMaskDC->SelectObject(bmpMask);

    plmage->Draw(
        pSrcDC,
        CRect(0, 0, w, h),
        CRect(0, 0, w, h)
    );

    COLORREF pozadina = pSrcDC->GetPixel(0, 0);
    pSrcDC->SetBkColor(pozadina);
    pMaskDC->BitBlt(0, 0, w, h, pSrcDC, 0, 0, SRCCOPY);

    pSrcDC->SetBkColor(RGB(0, 0, 0));
    pSrcDC->SetTextColor(RGB(255, 255, 255));
    pSrcDC->BitBlt(0, 0, w, h, pMaskDC, 0, 0, SRCAND);

    pDC->BitBlt(0, 0, w, h, pMaskDC, 0, 0, SRCAND);
    pDC->BitBlt(0, 0, w, h, pSrcDC, 0, 0, SRCPAINT);

    delete pSrcDC, pMaskDC;
}

```

```

void CMy18724FilipLakicevicView::OnDraw(CDC* pDC) {
    CLampaDoc* pDoc = GetDocument();
    ASSERT_VALID(pDoc);
    if (!pDoc)
        return;
    CRect rect;
    GetClientRect(&rect);
    CDC memDC;
    memDC.CreateCompatibleDC(pDC);
    CBitmap bmp;
    bmp.CreateCompatibleBitmap(pDC, rect.Width(), rect.Height());
    CBitmap* oldbmp = memDC.SelectObject(&bmp);
    int oldmode = memDC.SetGraphicsMode(GM_ADVANCED);
    XFORM form;
    memDC.GetWorldTransform(&form);
    DrawBackground(&memDC);
    //ukljucuju DrawBakcg ti je isto

    //odavde ti ide ako hoces da ti bude na polovini ekrana slika koju crtas
    //ali vodi red, jer to zavisi i koliko ti je background slika
    //znaci ovaj translate u zavisnosti od zadatka
    Translate(&memDC, rect.Width() / 2, rect.Height() - 200, false);

    //ovo pozivas u zavisnosti od fje, znaci moze i bez translate i sa bez
    DrawLampShadow(&memDC);
    DrawLamp(&memDC, false);
    Translate(&memDC, -rect.Width() / 2, -(rect.Height() - 200), false);

    //isto
    memDC.SetWorldTransform(&form);
    pDC->BitBlt(0, 0, rect.Width(), rect.Height(), &memDC, 0, 0,
SRCCOPY);
    memDC.SetGraphicsMode(oldmode);
    memDC.SelectObject(oldbmp);
}

```